

PortWise AI

**Container Congestion Predictor
& Port Operations Optimizer**

TEAM: CODECRAFTER



The Multi-Billion Dollar Bottleneck at Container Ports

Ships **wait days** at anchorage causing major supply chain delays

Port congestion **costs billions** in demurrage penalties and wasted fuel

Decisions are **fully reactive** managers respond only after congestion builds

BEFORE
SMOOTH OPERATIONS



AFTER
CONGESTION BUILDS



Intelligent, Proactive Port Operations

PortWise AI turns uncertainty into control with real-time visibility and 72-hour congestion foresight.

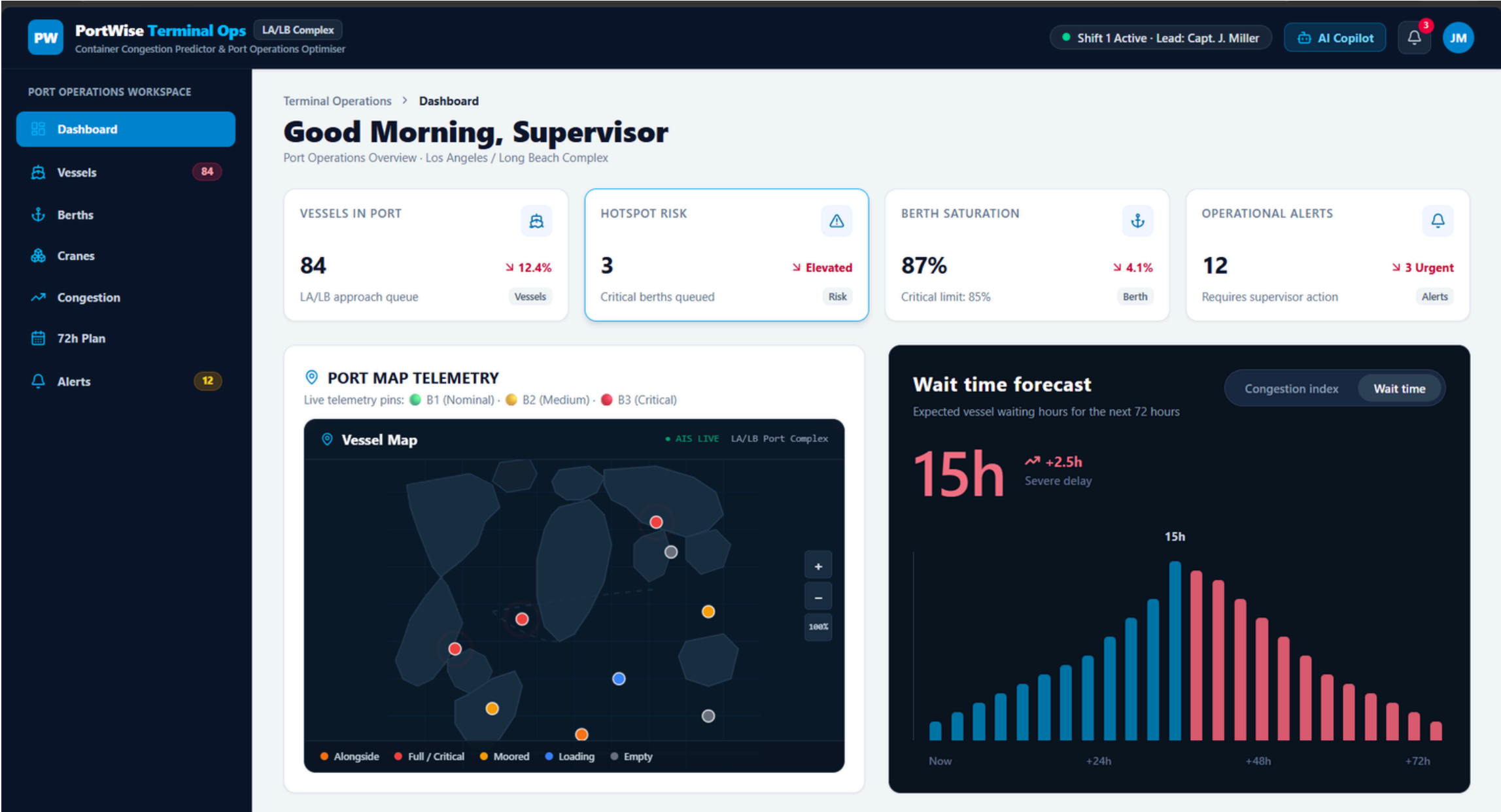
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Real-time AIS Vessel Tracking
- 02

72-Hour Congestion & Wait-Time Forecasting
- 03

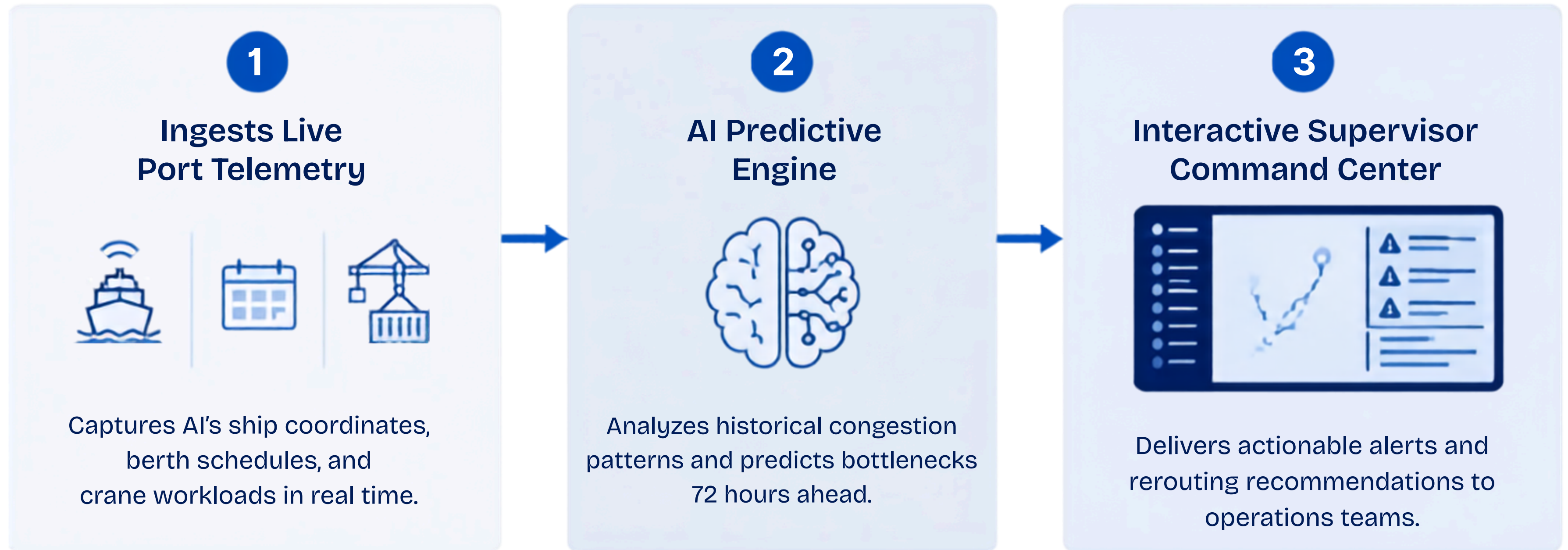
Smart Berth & Crane Saturation Balancing
- 04

Automated AI Mitigation & Rerouting



Behind the Scenes

How PortWise AI Operates



Built for Port Supervisors & Terminal Managers



Predictive Bar Charts

Hour-by-hour congestion
and waiting time forecasts



Interactive Port Map

Live visual telemetry
of berths, anchorage
queues, and ship status



Operational Alert Center

Instant conflict
notifications with
one-click AI mitigation



One-Click PDF Reports

Fast export of terminal
status for shift
handovers

Powered by IBM Bob & Intelligent AI Copilot

Instant copilot intelligence

Supervisors ask natural language questions and get clear answers instantly

Automated rerouting

AI detects berth conflicts in real time and suggests the best alternative docks

Dynamic shift planning

Optimizes crane crew shifts based on expected peak arrivals and workload

 **PortWise AI Copilot**
Real-time Port Intelligence

SUGGESTED OPERATIONAL QUERIES

Reroute Berth B-12 MSC Irina

Forecast peak congestion window

West Yard Stack crane boost

 Hello Capt. Miller. I am your PortWise AI Copilot. You can ask me to reassign berths, forecast queue delays, optimize crane shifts, or evaluate demurrage savings.

West Yard Stack crane boost

 I've initiated a load redistribution for the West Yard stack crane (YRD-WEST). To alleviate its critical load of 91%, some of its current tasks will be rerouted to nearby active cranes (e.g., YRD-NORTH and CRN-Q07). This will bring YRD-WEST back toward a safer operating range while maintaining overall throughput.

Reroute Berth B-12 MSC Irina

 Rerouting vessel MSC Irina to Berth B-12 has been initiated. We'll monitor the move and keep you updated on its progress.

which ship has highest weight

 The current data set only includes vessel statuses and does not contain any weight information for the ships, so I cannot determine which ship has the highest weight. If

Tangible Benefits for Port Authorities & Shipping Lines



35%

reduction in average
vessel waiting time
at anchorage.



25%

increase in berth
and crane utilization
efficiency.



\$300K+

estimated demurrage
savings per week for
busy terminals.



100%

proactive planning
replacing reactive
firefighting.

The Builders Behind PortWise AI

Team Member	Role
Jay Patel	Team Lead, Architecture & AI Workflow
Miral Patel	Full-Stack & Interactive UI/UX
Fenil Finava	Backend
Dhruvin Ramani	Port Simulation, Maps & Analytics